

Intelligent Agents

Outline

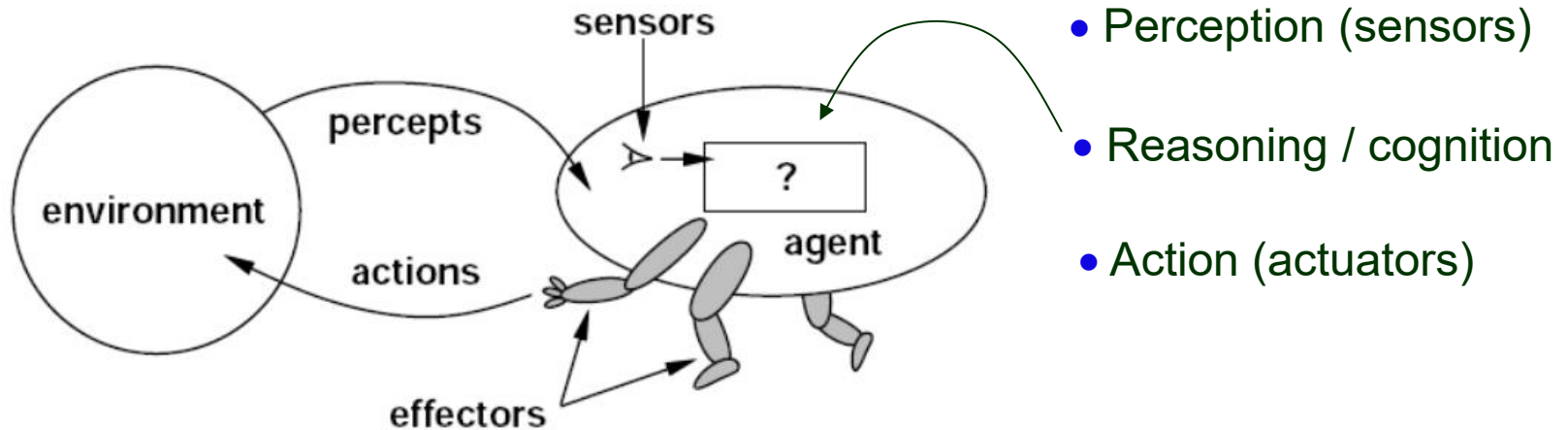
I. Intelligent agents

II. The nature of environments

* In part based on notes by Dr. [Jin Tian](#).

** Figures are from the [textbook site](#) (or drawn by the instructor) unless their sources are cited.

I. Intelligent Agents



Percept: perceptual inputs at any given instant.

Percept sequence: complete history of everything the agent has ever perceived.

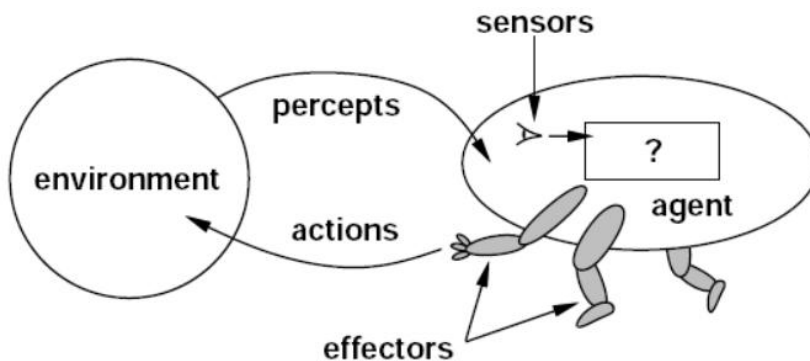
Agent function (behavior):

a percept sequence $\xrightarrow{\text{reasoning}}$ an action

Construction of the Agent Function

Tabulation?

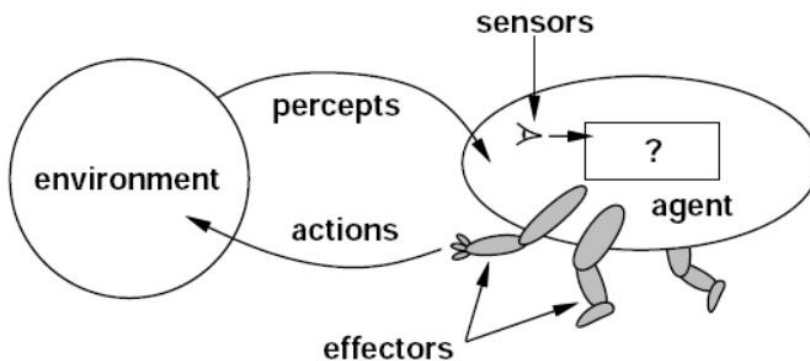
- ♠ Very large, if not infinite, table!
- ◆ Instead, implement the function internally by an agent program.
- ◆ The program runs on the agent's architecture to produce the function.



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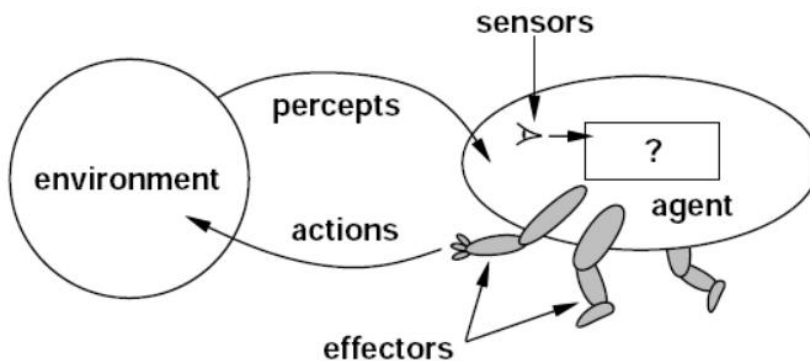


Agent = architecture + program

Construction of the Agent Function

Tabulation?

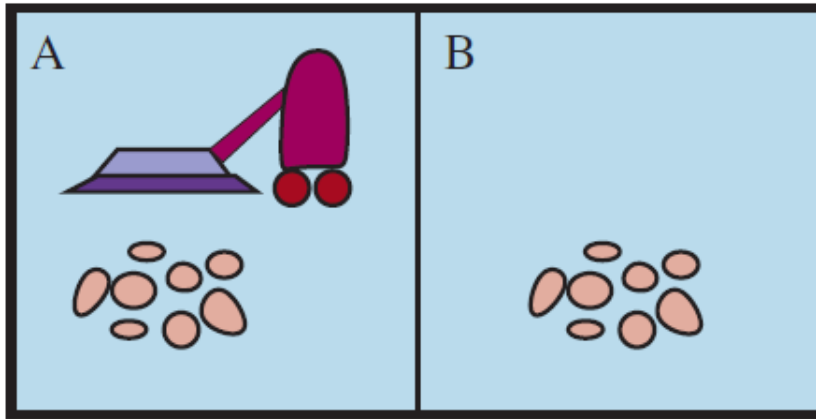
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Agent = architecture + program

- ◆ Abstract description vs concrete implementation!

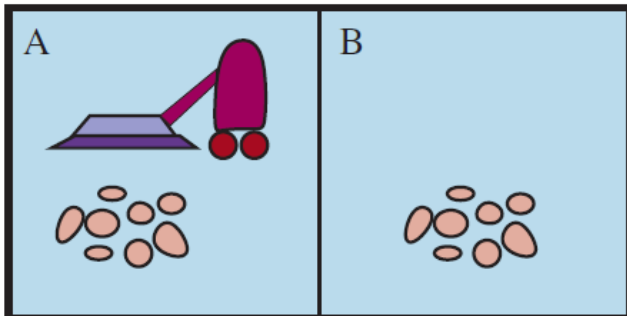
The Vacuum-Cleaner World



- *Environment*: squares *A* & *B*
- *Percepts*: [*A*, *Dirty*]
 - ↑
square the vacuum
cleaner is in
 - ↙
state of
the square
- *Actions*: *left*, *right*, *suck*, *nothing*

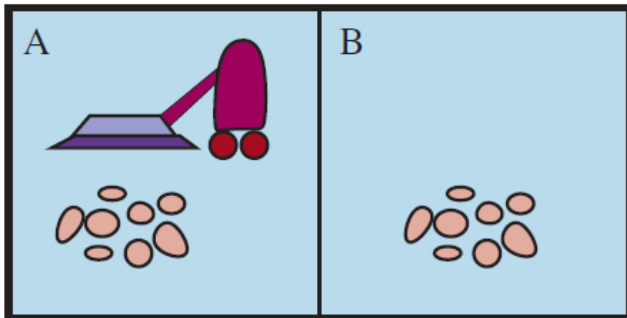
Agent Function

Percept sequence	Action
$[A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Dirty}]$	<i>Suck</i>
$[B, \textit{Clean}]$	<i>Left</i>
$[B, \textit{Dirty}]$	<i>Suck</i>
$[A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$
$[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$



Agent Function

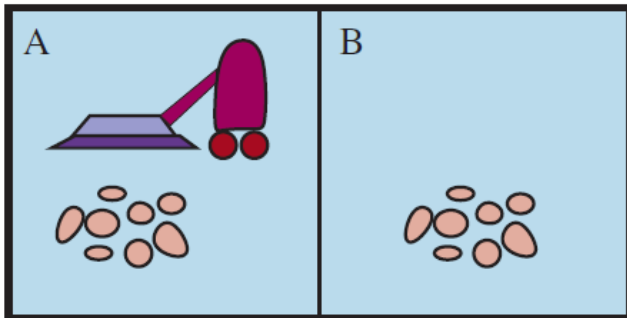
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$[B, \textit{Dirty}]$	<i>Suck</i>
$[A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$
$[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$



- Many ways to fill in the right column
- What is the right way?

Agent Function

Percept sequence	Action
<i>[A, Clean]</i>	<i>Right</i>
<i>[A, Dirty]</i>	<i>Suck</i>
<i>[B, Clean]</i>	<i>Left</i>
<i>[B, Dirty]</i>	<i>Suck</i>
<i>[A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>
<i>[A, Clean], [A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>

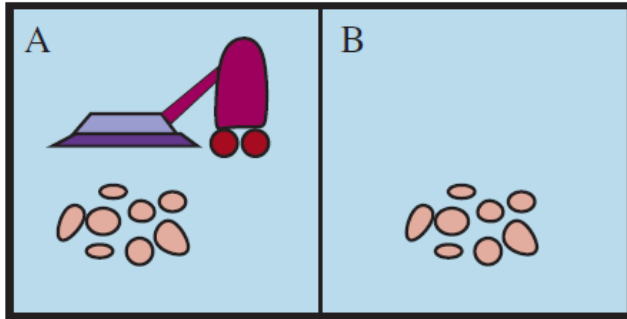


- Many ways to fill in the right column
- What is the right way?



Good/bad, intelligent/stupid?

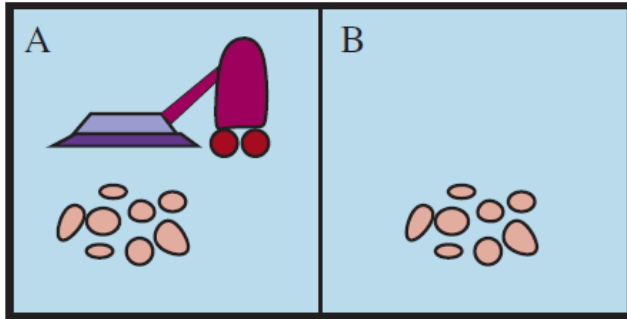
Rational Behavior?



if *status* == *Dirty* then return *Suck*
else if *location* == *A* then return *Right*
else if *location* == *B* then return *Left*

Is this agent rational?

Rational Behavior?

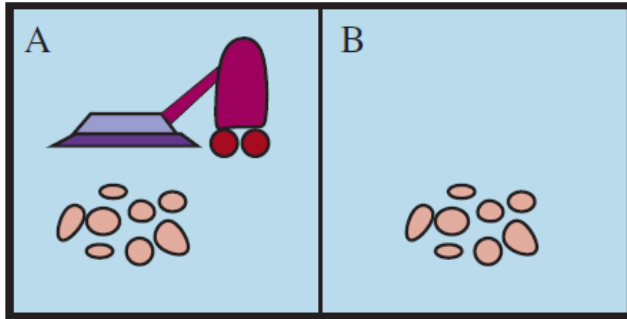


if *status* == *Dirty* then return *Suck*
else if *location* == *A* then return *Right*
else if *location* == *B* then return *Left*

Is this agent rational?

No, needless oscillation once all the dirt is cleaned up!

Rational Behavior?



if *status* == *Dirty* then return *Suck*
else if *location* == *A* then return *Right*
else if *location* == *B* then return *Left*

Is this agent rational?

No, needless oscillation once all the dirt is cleaned up!

↓ improve

Do nothing when all the squares are clean.

Rationality

What is rational depends on four things:

- ♦ performance measure defining the criterion of success
- ♦ prior knowledge of the environment
- ♦ performable actions by the agent
- ♦ perceptual sequence to date

A rational agent should select an action expected to maximize its performance measure.

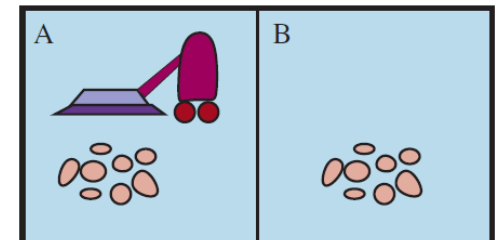
Performance Measure

- Awards one point for each clean square at each time step.

Meanwhile, assume

- known environment
- unknown dirt distribution and agent's initial location
- **only available** actions: *Left*, *Right*, and *Suck* (*Nothing* no longer an action)
- *Left* and *Right* having no effect if they would take the agent outside
- perfect sensing of location and dirt existence there

Percept sequence	Action
[A, Clean]	<i>Right</i>
[A, Dirty]	<i>Suck</i>
[B, Clean]	<i>Left</i>
[B, Dirty]	<i>Suck</i>
[A, Clean], [A, Clean]	<i>Right</i>
[A, Clean], [A, Dirty]	<i>Suck</i>
⋮	⋮
[A, Clean], [A, Clean], [A, Clean]	<i>Right</i>
[A, Clean], [A, Clean], [A, Dirty]	<i>Suck</i>
⋮	⋮



This agent is rational.

Omniscience vs Rationality

- An *omniscient agent* knows the actual outcome of its actions.

Impossible in reality!

- Rationality maximizes the expected performance.
 - ♦ Learn as much as it perceives.
 - ♦ Does not require omniscience.
- Perfection maximizes actual performance.

Rationality $\neq$ omniscience $\neq$ perfection

II. Task Environment

To design a rational agent, we must specify its task environment:

- performance measure
 - environment of the agent
 - agent's actuators and sensors
- 
- PEAS

Automated Taxi Driver

Its task environment in the PEAS description:

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

PEAS for Other Agents

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice

PEAS for Other Agents

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of satellite images, categorization results	
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice



Universal Robots
ActiNav autonomous
bin picking kit

Environment Properties

- Categorize task environments according to **properties**.



appropriate **families of techniques**
for agent implementation

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle Chess with a clock						
Poker Backgammon						
Taxi driving Medical diagnosis						
Image analysis Part-picking robot						
Refinery controller English tutor						

Environment Property 1

- **Fully observable** if the sensors can detect all aspects that are *relevant* to the choice of action.
vs. **partially observable**

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully					
Chess with a clock	Fully					
Poker	Partially					
Backgammon	Fully					
Taxi driving	Partially					
Medical diagnosis	Partially					
Image analysis	Fully					
Part-picking robot	Partially					
Refinery controller	Partially					
English tutor	Partially					

Environment Property 2

- Single-agent
vs. multiagent

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single				
Chess with a clock	Fully	Multi	← competitive			
Poker	Partially	Multi				
Backgammon	Fully	Multi				
Taxi driving	Partially	Multi	← cooperative			
Medical diagnosis	Partially	Single				
Image analysis	Fully	Single				
Part-picking robot	Partially	Single				
Refinery controller	Partially	Single				
English tutor	Partially	Multi				

Environment Property 3

- **Deterministic** if the next state of the environment is completely determined by the current state and the action executed by the agent.
- **vs. stochastic** otherwise (randomly determined or described by a random probability distribution).

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic			
Chess with a clock	Fully	Multi	Deterministic			
Poker	Partially	Multi	Stochastic			
Backgammon	Fully	Multi	Stochastic			
Taxi driving	Partially	Multi	Stochastic			
Medical diagnosis	Partially	Single	Stochastic			
Image analysis	Fully	Single	Deterministic			
Part-picking robot	Partially	Single	Stochastic			
Refinery controller	Partially	Single	Stochastic			
English tutor	Partially	Multi	Stochastic			

← unable to keep track of all the cards in opponents' hands; must be treated as nondeterministic

Environment Property 4

- **Episodic** if the agent's experience is divided into atomic episodes, among which one does not depend on the actions taken in previous ones.
- vs. **sequential** if the current decision could affect all future decisions.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential		
Chess with a clock	Fully	Multi	Deterministic	Sequential		
Poker	Partially	Multi	Stochastic	Sequential		
Backgammon	Fully	Multi	Stochastic	Sequential		
Taxi driving	Partially	Multi	Stochastic	Sequential		
Medical diagnosis	Partially	Single	Stochastic	Sequential		
Image analysis	Fully	Single	Deterministic	Episodic		
Part-picking robot	Partially	Single	Stochastic	Episodic		
Refinery controller	Partially	Single	Stochastic	Sequential		
English tutor	Partially	Multi	Stochastic	Sequential		

instantaneous actions
can have long-term
consequences.

Environment Property 5

- **Dynamic** if the environment can change while the agent is choosing an action.
- vs. **semidynamic** if the environment does not change but the agent's performance score does.
- vs. **static** otherwise.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	
Poker	Partially	Multi	Stochastic	Sequential	Static	
Backgammon	Fully	Multi	Stochastic	Sequential	Static	
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	
Image analysis	Fully	Single	Deterministic	Episodic	Semi	
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	

Environment Property 6

- Discrete
vs. continuous

The distinction applies to

- ♦ the environment's state
- ♦ the way time is handled
- ♦ the agent's percepts and actions

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete